

Заг. 1 $X_i \sim N(163; 5,6^2)$

1. $X_1 + \dots + X_{20} \sim N(163, \frac{5,6^2}{20})$

Забележете, че тук няма нужда от ЦГТ, т.к. работим с $N(\mu, \sigma^2)$!

$$P(N(163, \frac{5,6^2}{20}) > 165) = P(N(0,1) > \frac{2\sqrt{20}}{5,6})$$

$$\approx 1 - \Phi(1.60) \approx \underline{\underline{5.48\%}}$$

2. $P(X_i < 180) = P(N(0,1) < \frac{17}{5,6})$

$$\approx \Phi(3.04) \approx 0.9988$$

$$P(\text{поке } \perp \text{ наг } 180) = 1 - P(\text{всички наг } 180)$$

$$\approx 1 - 0.9988^{20} \approx \underline{\underline{2.37\%}}$$

Заг. 2 $\int_0^\infty \int_0^y c e^{-y} dx dy = 1 \Rightarrow$

$$1 = \int_0^\infty y e^{-y} dy = c [-e^{-y}(y+1)]_0^\infty = c, \text{ т.е.}$$

$$\underline{\underline{c = 1.}}$$

За $x > 0$: $f_x(x) = \int_x^\infty e^{-y} dy = e^{-x}$ и

знаем $X \sim \text{Exp}(1)$, откъдето $EX=1, DX=1$.
 Може и директно: $EX = \int_0^{\infty} x e^{-x} dx = 1$
 $EX^2 = \int_0^{\infty} x^2 e^{-x} dx = 2$, $DX = 1$

Аналогично, за $y > 0$: $f_Y(y) = \int_0^y e^{-y} dx = y e^{-y}$
 $EY = \int_0^{\infty} y^2 e^{-y} dy = 2$, $EY^2 = \int_0^{\infty} y^3 e^{-y} dy = 6$
 и знаем $DY = 2$

$$EXY = \int_0^{\infty} \int_0^y x y e^{-y} dx dy$$

$$= \int_0^{\infty} y e^{-y} \cdot \frac{y^2}{2} dy = \frac{1}{2} \int_0^{\infty} y^3 e^{-y} dy$$

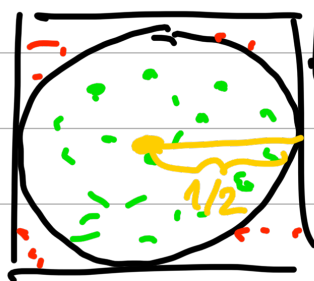
$$= 3.$$

Следователно, $\text{Cor}(X, Y) = \frac{3 - 1 \cdot 2}{\sqrt{1 \cdot 2}} = \frac{\sqrt{2}}{2}$.

2. $f_{X|Y}(x|y=1) = \frac{e^{-1}}{e^{-1}} = 1$ за $x \in (0, 1)$

и $E(X|Y=1) = \int_0^1 x \cdot 1 dx = \underline{\underline{\frac{1}{2}}}$.

заг. 3



Нека $X_i = \begin{cases} 1, & \text{ако точка } i \text{ е в кръга} \\ 0, & \text{ако не е.} \end{cases}$

$$P(X_i = 1) = \frac{\pi \cdot (\frac{1}{2})^2}{1} = \frac{\pi}{4}$$

Ако симулираме n точки, то заства, полагаме в кръг е $\frac{X_1 + \dots + X_n}{n}$.

$$X_i \sim \text{Ber}\left(\frac{\pi}{4}\right), \mathbb{E}X_i = \frac{\pi}{4} < \infty \text{ и от}$$

ЗГЗ: $\frac{X_1 + \dots + X_n}{n} \xrightarrow[n \rightarrow \infty]{\text{n.c.}} \frac{\pi}{4}$, т.е. възмож-

на оценка за π е:

$$\hat{\pi}_n = \frac{4(X_1 + \dots + X_n)}{n}$$

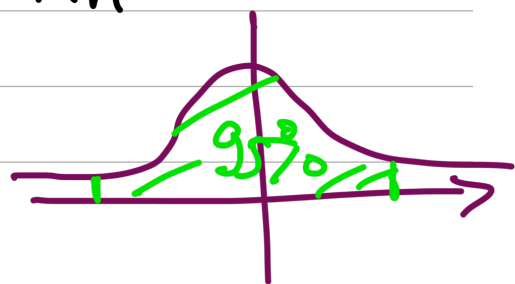
Избирам n , така че

$$P(-0.001 \leq \hat{\pi}_n - \pi \leq 0.001) = 95\%$$

Т.к. $DX_1 < \infty$, от ЦГТ: $\frac{X_1 + \dots + X_n}{n} - \frac{\pi}{4} \xrightarrow[n \rightarrow \infty]{\text{n.c.}} \frac{\sigma/\sqrt{n}}$,
където $\sigma^2 = \frac{\pi}{4} \left(1 - \frac{\pi}{4}\right)$

$$95\% = P\left(\frac{-0.001}{\sigma/\sqrt{n}} \leq \frac{\frac{X_1 + \dots + X_n}{n} - \frac{\pi}{4}}{\sigma/\sqrt{n}} \leq \frac{0.001}{\sigma/\sqrt{n}}\right)$$

$$= 1 - 2\Phi\left(-\frac{0.001}{\sigma/\sqrt{n}}\right)$$



и получаване $\Phi\left(-\frac{0,001}{4\sigma/\sqrt{n}}\right) = 0,025$

От таблицата: $\frac{0,001}{4\sigma/\sqrt{n}} \approx 1,96$

$$\Rightarrow n \approx \left(\frac{1,96 \cdot 4\sigma}{0,001}\right)^2$$

$$\approx \underline{\underline{3219}}$$

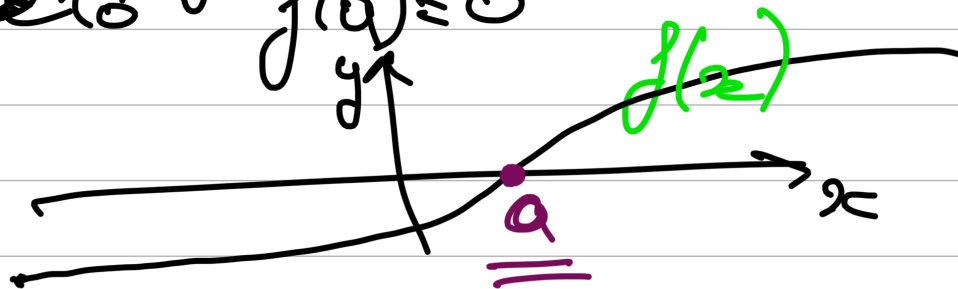
зад. 4 $f_X(x) = 1_{\{x \in [0,1]\}}$; $f_Y(y) = e^{-y} 1_{\{y > 0\}}$

1. Да върко е. Дорманко,

$$P(X \leq a) - P(Y \geq a) =: f(a) \text{ е}$$

непрекъсната, $f(-\infty) = -1$, $f(\infty) = 1$

и по теоремата на Болцано-Вейерштрас (за средната ст-т), съществува a за което $f(a) = 0$



$$2. \begin{cases} Z = \frac{X}{Y} \\ W = Y \end{cases} \Rightarrow \begin{cases} X = ZW \\ Y = W \end{cases} \text{ и } |J| = \begin{vmatrix} W & Z \\ 0 & 1 \end{vmatrix} = W$$

$$f_{Z,W}(z,w) = 1_{\{ZW \in [0,1], w > 0\}} e^{-w} \cdot w$$

$$= 1_{\{z > 0, \frac{1}{z} > w > 0\}} e^{-w} \cdot w$$

и знаем за $z > 0$: $f_Z(z) = \int_0^{\frac{1}{z}} e^{-w} \cdot w \, dw$
 $= \underline{1 - e^{-1/z} \left(\frac{1}{z} + 1 \right)}$ и 0 иначе.

$$\begin{cases} Z = \frac{Y}{X} \\ W = X \end{cases} \Rightarrow \begin{cases} X = W \\ Y = ZW \end{cases} \text{ и } |J| = \begin{vmatrix} 0 & 1 \\ W & Z \end{vmatrix} = -W.$$

Следователно $f_{Z,W}(z,w) = 1_{\{W \in [0,1], z > 0\}} e^{-ZW} \cdot w$

и в такъв случай, за $z > 0$

$$f_Z(z) = \int_0^1 w e^{-zw} \, dw = \frac{1}{z^2} \int_0^z t e^{-t} \, dt$$

$$= \underline{\frac{1}{z^2} \left(1 - e^{-z} (z+1) \right)}$$

и 0 за $z \leq 0$.

* За бонуса : Ако $Z = \frac{X}{Y}$, $W = \frac{Y}{X}$,

то $f_Z(z) \sim 1 - (1 - \frac{1}{z})(\frac{1}{z} + 1) = \frac{1}{z^2}$

и следователно $\mathbb{E} Z = \int_0^{\infty} z f_Z(z) dz = \infty$

т.к. $\int_0^{\infty} \frac{1}{x} dx = \infty$.

За W : $f_W(x) \sim \frac{1}{x^2}$ и аналогично

$\mathbb{E} W = \int_0^{\infty} x f_W(x) dx = \infty$.